Clean with pause · 80 kg · 65 %

Bertrand Cadet · 25 Oct 2026 at 17:10 · Made

97 / 100 · technique score (bar path)

Criterion	Points
Forward in the second pull	28 / 30
Catch position	20 / 20
First pull backward	15 / 15
Drop under the bar	15 / 15
Bar tilt	9 / 10
Bar path type	10 / 10

Bar path



Fix first

- **The bar goes forward in the second pull**
  It moves 5 cm away after hip contact. Keep the arms long and relaxed, push the floor all the way, and bring the bar back toward you as you go under. Goal ≤ 5 cm: decrease by 0,4 cm.
- **Bar tilted**
  It tilts up to 3,5°, right side higher. Check the symmetry of your grip and leg drive. Goal ≤ 3 °: reduce by 0,5 °.

Tips by metric

- **Forward in the second pull ↓**
  5,4 cm · goal ≤ 5 cm · decrease by 0,4 cm

  After hip contact, the bar drifts away from the body: you have to chase it forward, which costs height and causes misses in front. Common causes: hips banging the bar, arms pulling too early, extending forward.

  Instructions
  - Arms long and relaxed until the extension is complete.
  - The bar slides along the thighs and hip crease: brush it, don't bang it.
  - Extend up, not forward.
  - After the extension, bring the bar back toward you: elbows high and out.

  Exercises
  - Clean from the hang (bar at the hip).
  - High pull, bar in contact with the body.
  - Muscle Clean (no dip under the bar).
  - Slow segmented pulls mid-thigh → hip.

- **Max tilt ↓**

  3,5 ° · goal ≤ 3 ° · reduce by 0,5 °

  One side of the bar rises higher than the other: asymmetry in grip, leg drive or pelvic rotation.

  Instructions

  - Symmetrical grip: hands equally spaced from the bar markings.
  - Feet symmetrical at the start.
  - Push the floor evenly with both feet.

  Exercises

  - Slow pulls facing a mirror or filmed from the front.
  - Bulgarian split squats and lunges to balance both legs.

Bar path

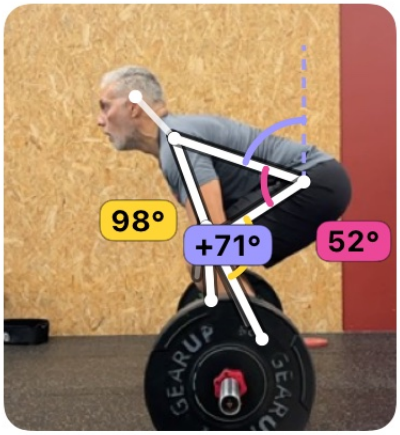
Measurement	Value	Goal · to do
First pull backward bar brought back toward the athlete	5,4 cm	goal ≥ 3 cm OK
Forward in the second pull after the rearmost point	5,4 cm	goal ≤ 5 cm ↓ decrease by 0,4 cm
Back (loop) from the foremost point to the catch	9,4 cm	
Catch position relative to the start line	-9,4 cm	goal ≤ 2 cm OK
Max height 65 % of height	119 cm	
Height at peak velocity point V, -0,2 cm from the start line	100 cm	
Height without arm pull bar released at peak velocity: h(V) + v²/2g	111 cm	
Arm contribution actual max height – height without arm pull	+8,4 cm	
Catch height 49 % of height	90 cm	
Drop under the bar max height – catch	29,5 cm	goal ≤ 30 cm OK
Bar path type comes back toward the athlete and stays behind the line	Type 2	

Velocity and rhythm (sensor)

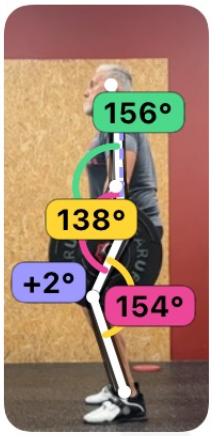
Peak velocity Second pull	1,49 m/s	
Velocity at end of first pull before the transition	-0,00 m/s	
Mean velocity, first pull excluding pause	0,17 m/s	
Mean velocity, second pull from the transition to peak velocity	0,74 m/s	
Peak acceleration, first pull 0,13 g on top of gravity	1,2 m/s²	
Peak acceleration, second pull 0,87 g on top of gravity	8,5 m/s²	
First pull duration excluding pause	2,93 s	
Pauses (3) plate height · duration	31 cm · 2,2 s 56 cm · 2,0 s 80 cm · 1,1 s	

Velocity and rhythm (sensor)		
Second pull duration start from the last pause	0,28 s	
Turnover from peak velocity to the catch	0,50 s	
Stabilization in the catch from the catch to the start of the recovery	0,23 s	
Recovery standing position not reached before the end	—	
Max tilt right side higher	3,5°	goal ≤ 3 ° ↓ reduce by 0,5 °
Max horizontal rotation one side is ahead of the other	1,5°	

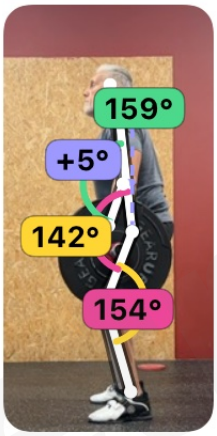
Posture: positions at key moments



Lift-off



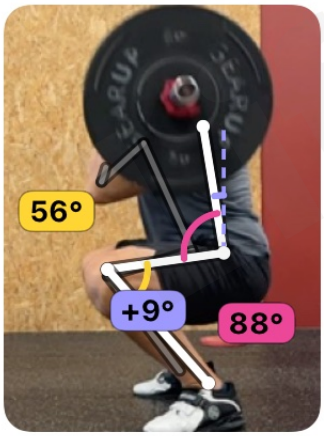
End of 1st pull



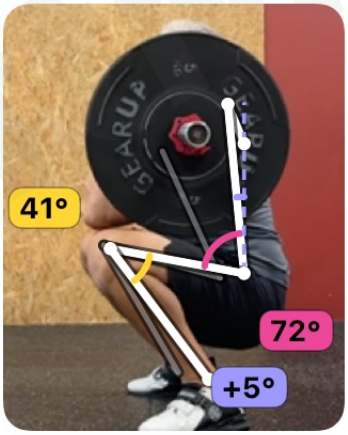
Transition



Peak velocity



Catch



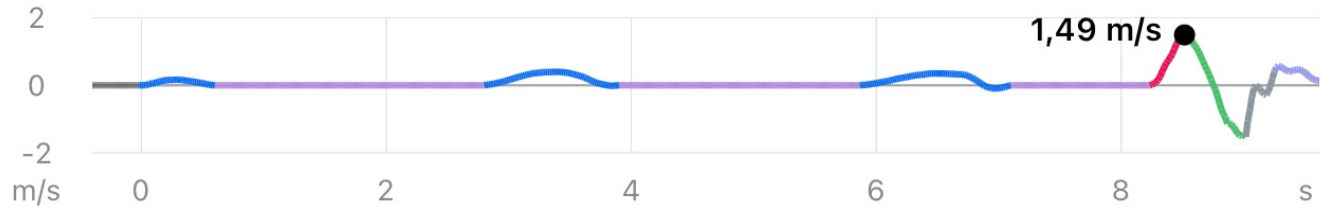
Bottom

Posture (angles read from the video)				
	Knee	Hip	Trunk	Elbow
Lift-off	98°	52°	+71°	161°
End of 1st pull	138°	154°	+2°	156°
Transition	142°	154°	+5°	159°
Peak velocity	151°	170°	−6°	142°
Catch	56°	88°	+9°	—
Bottom	41°	72°	+5°	—

Drawings: yellow = knee, pink = hip, purple = trunk relative to vertical, green = elbow (when bent). Knee, hip and elbow 180° = straight; trunk 0° = vertical, + leaning forward, – leaning back. 2D side-on measurement, accurate to a few degrees.

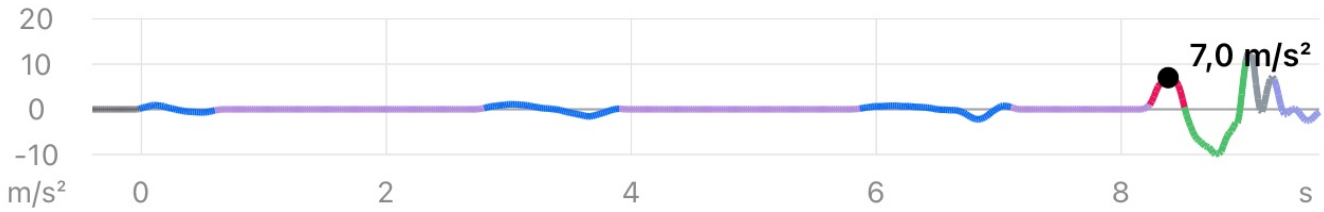
LiftMetrics

Velocity / time (m/s)

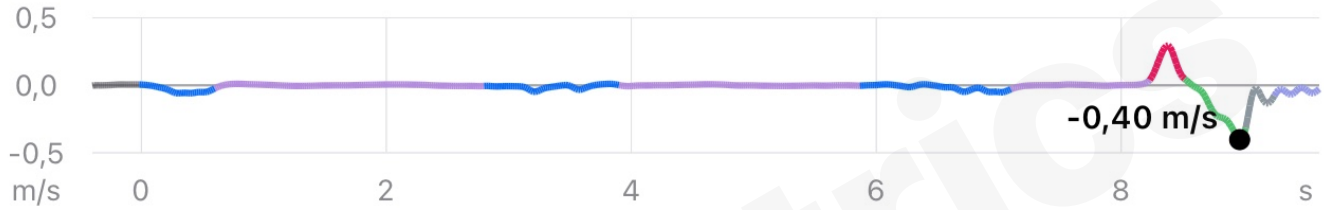


Acceleration / time (m/s²)

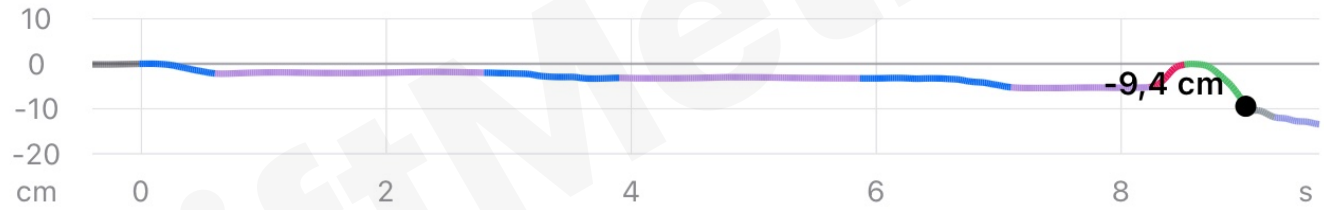
smoothed video curve; peak measured by the sensor: 8,5 m/s²



Horizontal velocity / time (m/s)



Horizontal displacement / time (cm)



Power / time (W)



LiftMetrics

How to read these numbers

Goal · to do	Target range for each measurement (adjustable in Settings › Goals). Green = within range · orange = outside but within tolerance · red = beyond. “increase / decrease by X” = gap to close to get back in range.
Score /100	Bar path quality, independent of load. Points: second-pull forward displacement 30, catch position 20, first-pull backward displacement 15, drop under the bar 15, bar tilt 10, bar path type 10. Each criterion earns full points within its range, half at the tolerance limit, then decreases to 0. An unmeasured criterion is left out of the calculation. Color: green from 80, orange from 60 to 79, red below 60.
Green / orange / red colors	Green = on target · orange = slightly outside (within tolerance) · red = to correct. The ranges for each measurement are given below.
Distance sign	Horizontal distances are measured from the start line (vertical through the bar at lift-off): + = in front of the line (away from the athlete), – = behind (toward the athlete).
Bar path colors	Blue = first pull · yellow = transition · pink-red = second pull · green = turnover · gray = catch · mauve = recovery · purple = pause. The white dot marks peak velocity. Velocity graph: same phase colors.
Backward (first pull)	Distance the bar comes back toward the athlete between lift-off and its rearmost point. Clean: goal ≥ 3 cm, orange up to 3 cm outside, red beyond.
Forward (second pull)	Distance the bar travels forward between its rearmost and its foremost point (bar “thrown” away from the body). Clean: goal ≤ 5 cm, orange up to 3 cm outside, red beyond.
Catch	Horizontal position of the bar at the catch relative to the start line. Clean: goal ≤ 2 cm, orange up to 3 cm outside, red beyond.
Drop	Drop under the bar: max height – catch height (the bar comes down before being caught). Clean: goal ≤ 30 cm, orange up to 8 cm outside, red beyond.
Arms (arm contribution)	Actual max height – height the bar would have reached if released at peak velocity ($h(V) + v^2/2g$). Measures what the arm pull adds after the extension.
Bar path type	1: the bar comes back toward the athlete then crosses in front of the line again · 2: it comes back and stays behind the line · 3: it doesn't come back toward the athlete (penalized in the score).
Tilt	Maximum bar tilt measured by the sensor (one side higher than the other), in degrees. Clean: goal ≤ 3 °, orange up to 2 ° outside, red beyond.

LiftMetrics